Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE 227: Data Communication

Batch: 22nd

Date: 22/12/23

Mid-Term Examination

Semester: Fall 2023

Time: 1 Hour 30 Minutes

Total Marks : 30

Answer any **five** questions: $5 \times 6 = 30$

1. (a) Explain the components of data communication. 3
   (b) Define the fundamental characteristics of effective data communication. 3

2. (a) Draw and explain the Maturity levels of an RFC. 3
   (b) Explain the difference between the duties of the IETF and IRTF. 3

3. (a) Illustrate layered architecture of TCP/IP protocol suite. 3
   (b) Show the encapsulation and decapsulation process in TCP/IP model. 3

4. (a) Describe the phases in periodic analog signals. 3
   (b) Show the addressing in TCP/IP protocol suite. 3

5. (a) Explain the modulation of a digital signal for transmission on a band pass channel. 3
   (b) Name the three types of transmission impairment. Explain. 3

6. (a) Explain the bandwidth-delay product. 3
   (b) Define Bandwidth, Throughput and Latency. 3

7. (a) Distinguish between a signal element and a data element. 3
   (b) Define a DC component and its effect on digital transmission. 3

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 225: Digital Logic Design

Batch: 22nd | Semester: Fall 2023

Date: 15/12/2023 | **Mid-Term Examination** | Time: 1 Hour 30 minutes | Total Marks: 30

Answer any five questions of the following. $5 \times 6 = 30$

1.		Convert the following numbers:	
	(a)	$(127.54)_8 = (?)_{10}$	3
	(b)	$(8A9B)_{12} = (?)_{18}$	3
2.	(a)	Write the difference between Combinational and Sequential logic.	3
	(b)	Apply Boolean identities and simplify this function $F(A, B, C) = (A+B)(A+C)$.	3
3.		Write the Additive, Multiplicative, Distributive, and Associative laws of Boolean Algebra.	6
4.	(a)	Implement three Basic Gates using any Universal Gate.	3
	(b)	Explain Exclusive Gates with the necessary equations and tables. Draw the circuit using only basic gates.	3
5.	(a)	Explain De Morgan's Theorem.	2
	(b)	How can you draw the 8-bit adder diagram using only one half adder?	4
6.	(a)	Draw the diagram of the Synchronous Sequential Circuit	2
	(b)	How can we build a Full Adder using the concept of half adder? Draw the required figure in your answer.	4
7.	(a)	Explain JK flip-flop with the necessary circuit and table.	3
	(b)	Is it possible to make a T flip-flop by modifying JK? Justify your answer with the necessary figures.	3

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

ENS221: Environmental Science

Mid-term Examination

Semester: Fall 2023

Time: 1 Hour 30 Minutes

Total Marks: 30

Date: 23/12/ 23

Answer any **three** of the following questions: **3×10 = 30**

1. a) Define environmental science with major components. 2

   b) 'Environmental Science is a multi-subjective approach'. Explain. 4

   c) 'Human being adjust with nature'. Justify it. 4

2. a) The photosynthesis process: $CO_2 + H_2O \rightarrow C_6H_{12}O_6 + O_2$. It is helps to maintain a relationship between plant and animal. Justify it. 2.5

   b) Give a description of hydrologic cycle in your own words. 4.5

   c) Write the sources of main three greenhouse gases. 3

3. a) What do you think about the main idea of 'Sustainable Development'? Could you relate between the objectives of sustainable development and the 'Three Zero Theory'? Explain. 6

   b) 'Sustainable development protects the natural resources'. Clarify it. 4

4. a) Sketch the vertical structure of atmosphere and elucidate stratosphere and thermosphere of it. 4

   b) 'Bangladesh is a flood prone country'. Explain. 3

   c) 'The sustainable agriculture helps to ensure the food security'. Justify it. 3

5. a) Delineate Coastal Region with figure. 2

   b) The number of cyclcone is increaing day by day. Provide your logic for supporting this statement. 2

   c) Write the mechanism of storm surge. Enumerate the effects of storm surge in the coastal region. 6

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 223: Object Oriented Programming in Java

Batch: 22nd | Semester: Fall 2023
Time: 1 Hour 30 Minutes
Date: 09/12/2023 **Mid-Term Examination** Total Marks: 30

Answer any **five** questions of the followings. $5 \times 6 = 30$

1. (a) Briefly explain the functionalities of **Java Virtual Machine (JVM)** and **Java Runtime Engine (JRE).** 3
   (b) Write a program in Java to find the **maximum** and **minimum** element in an array. 3

2. (a) Write the following code using **for** loop and show the output. 3

```
int i=1;
do{
   int j=1;
   do{
      System.out.print(i);
      j++;
   }while(j<=i);
   System.out.println();
   i++;
}while(i<=5);
```

   (b) Write a Java program for the following series- 3

$$1^2 - 2^2 + 3^2 - 4^2 - \ldots +/- n^2$$

3. (a) Briefly explain any three features that differentiate Object-Oriented Programming from Structured Programming Languages. 2
   (b) Explain the **encapsulation** of Object-Oriented Programming with an appropriate example. How does encapsulation ensure the data security of class members? 4

4. (a) Consider a class name is 'MyClass'. Create two objects 'ob1' and 'ob2' in Java main class for the given class. 2
   (b) Write a program in Java for **Sequential Search** of an array. 4

5. (a) What is a constructor in Java? What is the prime difference between constructor and method? 2
   (b) Write a program in Java for **Fibonacci Series.** 4

6. (a) Briefly explain the relation between **class** and **object** in Java. 2

(b) Write a method that takes an array of integers as an argument and returns a value based on the sums of the even and odd numbers in the array. Let X = the sum of the odd numbers in the array and let Y = the sum of the even numbers. The method should return X – Y. 4

7. (a) Briefly explain 'this' keyword in Java with example. 2

(b) An array with an odd number of elements is said to be centered if all elements **(except the middle one)** are strictly greater than the value of the middle element. Note that only arrays with an odd number of elements have a middle element. Write a method that accepts an integer array and returns 1 if it is a centered array, otherwise it returns 0. 4

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

CSE221: Computer Algorithms

Batch: 22nd — Semester: Fall 2023

Time: 1 Hour 30 minutes

Date: 08/12/2023 — **Mid-Term Examination** — Total Marks: 30

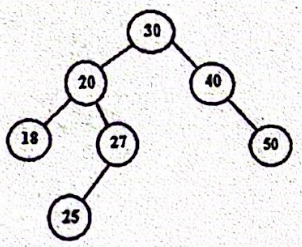
Answer any **three** questions of the followings. $10 \times 3 = 30$

1. (a) Explain the criteria of an algorithm. 2
   (b) What does it means by analysis of algorithm? Explain three asymptotic notations used to compute the efficiency of an algorithm. 4
   (c) Find the complexity from the following recurrence: 4

$$T(n) = \begin{cases} 1 & if\ n = 0 \\ T(n-1) + n & n > 0 \end{cases}$$

2. (a) Describe the procedures of Divide and Conquer method in algorithm. 3
   (b) Suppose CSE221 is a list of 12 number as follows: 3
   57 32 61 59 37 62 78 12 70 40 35 23
   Use the Quicksort Algorithm to find the position of the first number 57.
   (c) Write the Merge Sort Algorithm using Divide and Conquer method. 4

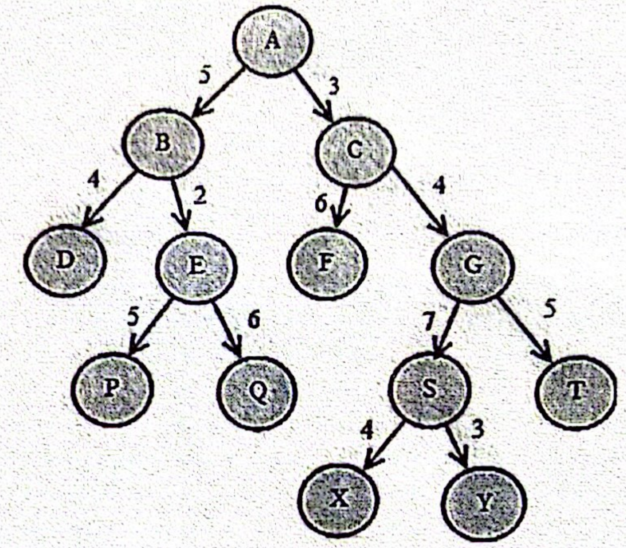
3. (a) Define **AVL** tree. Insert 60 and 70 into the following AVL tree. 3



   (b) State the recursive Mini-Max algorithm. 3
   (c) Define Convex Hull. Write the procedure to generate convex hull using QuickHull Algorithm. 4

4. (a) What does it mean by Greedy Method? Explain its components. 3
   (b) Consider the following instance of knapsack problem: 3
   Number of objects, $n = 6$;
   Capacity of knapsack, $m = 14$;
   Profits, $\{p_1, p_2, p_3, p_4, p_5, p_6\} = \{10, 5, 15, 7, 21, 18\}$ *and*
   Weights, $\{w_1, w_2, w_3, w_4, w_5, w_6\} = \{3, 2, 4, 4, 4, 5\}$.
   Calculate the optimal solution.

(c) Consider the tolerance level of the following network is 12. Now split the network with the help of Greedy Method.



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 227: Data Communication

Batch: 22nd | | Semester: Fall 2023
Date: 23/03/2024 | **Final Examination** | Time: 2 Hours
Total Marks : 40

Answer any **five** questions: $5 \times 8 = 40$

1. (a) Explain three different sampling methods for PCM. 4
   (b) List three different techniques in serial transmission and explain the differences. 4

2. (a) Draw and explain amplitude modulation in analog-to-analog conversion. 4
   (b) Which of the four digital-to-analog conversion techniques (ASK, FSK, PSK or QAM) is the most susceptible to noise? Defend your answer. 4

3. (a) List and explain all the categories of multiplexing. 4
   (b) Illustrate three strategies of time division multiplexing. 4

4. (a) Explain spared spectrum and its goal. 4
   (b) Draw and explain digital signal service (Digital hierarchy). 4

5. (a) How does sky propagation differ from line-of-sight propagation? 4
   (b) List the differences between omnidirectional waves and unidirectional waves. 4

6. (a) Draw a table for categories of UTP cables. 4
   (b) Write the advantages and disadvantages of fiber optic cable. 4

7. (a) Explain packet switching network. 4
   (b) Describe three phases of circuit-switched network. 4

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

ENS221: Environmental Science — Semester: Fall 2023

Final Examination — Time: 2 Hours

Date: 19/03/24 — Total Marks: 40

Answer any **four** of the following questions: **4×10 = 40**

1.	a)	'Environmental science covers four major areas'. Explain.	3
	b)	Write the significance of atmosphere in your life.	3
	c)	How you can apply your own ideas to mitigate the drought?	4
2.	a)	Define sedimentation process.	2
	b)	Illustrate the sedimentation processes with figures in the perspective of Bangladesh.	8
3.	a)	What do you think about the main idea of 'Green Tourism'? Explain the major indicators of green tourism.	6
	b)	Analyze the Sustainable Tourism in the light of EEE or PPP Concept in the context of Bangladesh.	4
4.	a)	Sketch the process of eutrophication. Explain the process in your own words.	4
	b)	Mention the impacts of two major floods in Bangladesh.	2
	c)	How you can apply recycle, reuse, reduce and reject in the context of waste management in your house?	4
5.	a)	The capacity building and institutional strengthening is one of the most important pillars among six pillars of Bangladesh Climate Change Strategy and Action Plan (BCCSAP). What do you think? Justify it. State the three programs of it.	6
	b)	Give some suggestions for preventing the damage of flash flood damage in Bangladesh.	4
6.		Write short note (any two)	10

i) National Adaptation Programme of Action 2005

ii) Environmental Conservation Act 1995

iii) Bangladesh Climate Change Strategy and Action Plan 20010

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE 225: Digital Logic Design

Batch: 22nd
Semester: Fall 2023
Time: 2 Hour
Date: 16/03/2024
Final Examination
Total Marks: 40

Answer any **five** questions of the following. **5 × 8 = 40**

1	(a)	Define Cache Memory. Write the Difference Between PROM and EPROM.	4
	(b)	What is memory word? Explain computer-level hierarchy.	4
2.	(a)	State Language Processing System? Show the steps of this system with proper naming.	4
	(b)	Draw and explain the Von Neumann model architecture.	4
3.	(a)	Explain fetch-execute cycle. Write differences between address and contents.	4
	(b)	Simplify the following expression in SOP and POS form using K-Map: **F(A, B, C, D) = AB'C + A'BCD' + ABC'D + ABC.**	4
4.	(a)	A processor communicates with memory and I/O devices by using signals. Signals are travelled along set of wires or connections. We called them by a specific term. Explain this with proper figure.	4
	(b)	Show a comparative analysis between serial and parallel ports. Write the differences between SOP and POS.	4
5.	(a)	What is auxiliary or backup memory? Give two examples of this kind of memory with features.	3
	(b)	Minimize the following Boolean function using sum of products (SOP) **f(a,b,c,d) = $\sum$m(3,7,11,12,13,14,15)**	5
6.	(a)	How many 3 to 8 line decoders do we need to implement a 4 to 16 line decoder? Draw the necessary diagram to justify your answer.	4
	(b)	Expand **A + BC' + ABD' + ABCD** to minterms and maxterms.	4
7.	(a)	Write the advantages for both High-level and Assembly languages.	4
	(b)	How can we implement 16x1 line Multiplexer using only lower-order Multiplexers?	4

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 223: Object Oriented Programming in Java

Batch: 22nd | Semester: Fall 2023
Time: 2 Hours
Date: 12/03/2024 | **Final Examination** | Total Marks: 40

Answer any **five** questions of the followings. $5 \times 8 = 40$

1. (a) Draw the life-cycle of Thread. Discuss "runnable" and "waiting" states of Thread in brief. 4

   (b) You need to write a method **editStudentDetails( )** that takes name, age, and CGPA as input. You will modify these values inside the method, and these changed values should be available outside the method. Show in the **main( )** body of the program how do you call this method and access those changed values. 4

2. (a) What are the differences between declaring a variable in a class with "protected" access modifier rather than no modifier? 2

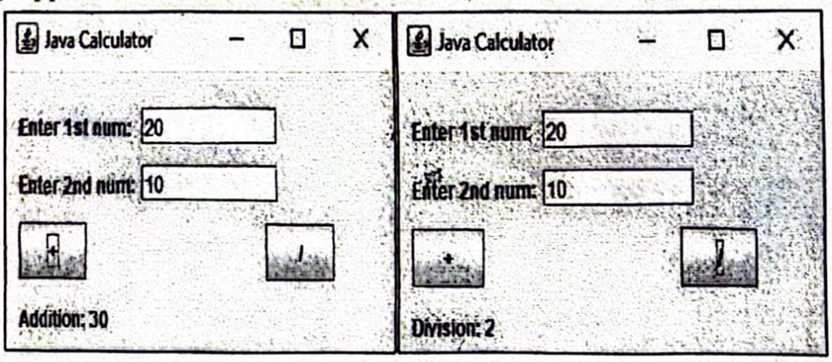
   (b) Consider the following code snippet: 6

```
String sl = "It";
String s2 = "was";
String s3 = sl + " " + s2;
s2 += " roses, ";
s3 = s3 + s2 + "roses all the way.";
System.out.println( s3);
```

   What will be the output of this code snippet? How many String objects will be created in this code snippet?

3. (a) What is the use of an *import* declaration in a Java program? 2

   (b) Write a Java program to display the following interface using AWT, Swing/Applet class with **ActionListener**. 6



4. (a) Explain the life-cycle of Applet in Java. 3

(b) What are the differences between Java applications and applets? Write the Java code to create a simple applet which draws the string "**Welcome to Java**". What code we need to write to embed this applet in an HTML document? 5

5. (a) Write the basic structure of Exception Handling block. 2

(b) Write a program in Java where each iteration of the **for** loop obtains two random integers. Those two integers are divided by each other, and the result is used to divide the value **12345**. The final result is put into **a**. If either division operation causes a **divide-by-zero** error, it is caught, the value set to zero, and the program continues. 6

6. (a) Why Do We Need Enumeration in Java? Write the basic syntax of Enumeration. 3

(b) Write a Java program to create an enum called "**DaysOfWeek**" representing the days of the week. 5

7. (a) Define Inheritance. How is inheritance used in polymorphism? 3

(b) Write the operation names of files in Java. Write a program in Java to check whether a given number is a palindrome or not. 5

temp = n

n =

HAMDARD UNIVERSITY BANGLADESH

In Pursuit of Knowledge

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE221: Computer Algorithms

Batch: 22nd — Semester: Fall 2023
Time: 2 Hours
Date: 10/03/2024 — **Final Examination** — Total Marks: 40

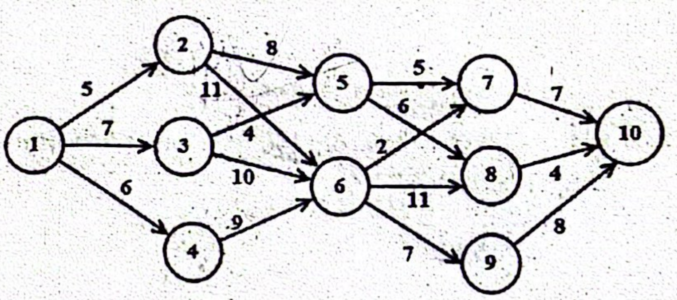
Answer any **five** questions of the followings. $5 \times 8 = 40$

1. (a) Write the applications and limitations of Greedy Algorithm. 4

   (b) Let you have four slots and seven jobs with the following profits and deadlines respectively: 4
   $\{p_1, p_2, p_3, p_4, p_5, p_6, p_7\} = \{20, 17, 12, 10, 9, 8, 6\}$ *and*
   $\{d_1, d_2, d_3, d_4, d_5, d_6, d_7\} = \{3, 1, 2, 1, 3, 4, 3\}$
   If each job takes one unit of time to execute, find the optimal solution using Greedy Algorithm.

2. (a) Write the pros and cons of bottom-up approach in problem solving. 2

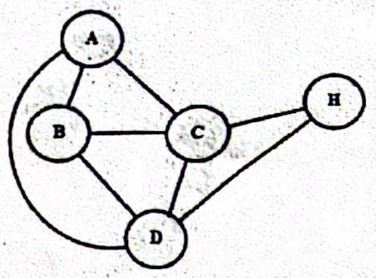
   (b) Find the minimum cost from node 1 to 10 for the following multistage graph: 6



3. (a) Differentiate between Dynamic programming and Backtracking. 3

   (b) Consider the following instance of the 0-1 knapsack problem: 5
   Number of items, $n = 4$,
   Capacity, $W = 8$, $S^1 = S^0 \cup S^1$
   Profits $\{p_1, p_2, p_3, p_4\} = \{5, 6, 7, 11\}$ *and*
   Weights $\{w_1, w_2, w_3, w_4\} = \{2, 3, 3, 5\}$
   Calculate the optimal solution using dynamic programming.

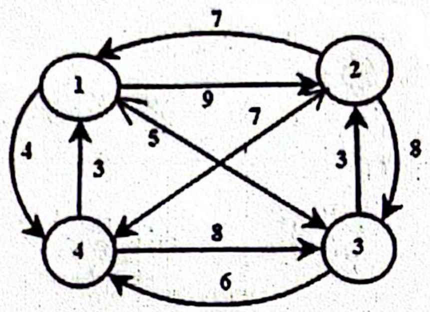
4. (a) Find two Hamiltonian circuit for the following graph using Backtracking method: 3



   (b) Find three solutions for **n-queens** problem using backtracking when n = 5. 5

5. (a) Write the advantages and disadvantages of backtracking. 3

(b) Solve Travelling Salesman problem with the help of dynamic programming for the following situation: 5



6. (a) Differentiate between **LIFO** Branch-and-Bound and **FIFO** Branch-and-Bound technique. 3

(b) Apply branch-and-bound technique to solve 15-puzzle problem for the following organizations: 5

1	2	3	
5	6	7	4
9	10	12	8
13	14	11	15

7. (a) Write short note on Lower Bound Theory.

(b) Define decision tree. Prove the given theorem: Any decision tree that can sort n elements must have **Ω(nlogn)** height.